

Geo-informatics applications in Mining

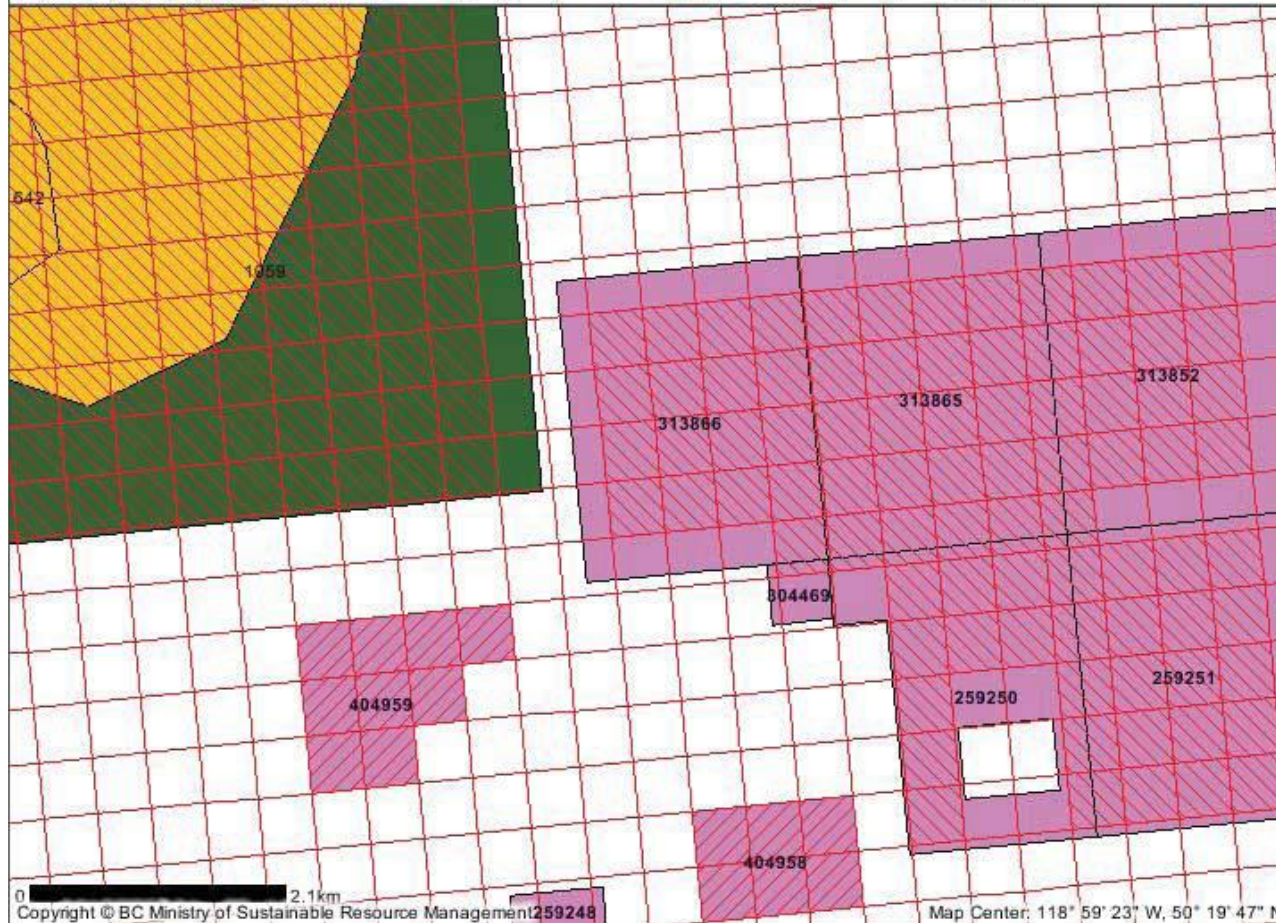
- Mineral exploration geoscientists use diverse types of datasets to search for new economic deposits.
- Data sources vary from geologic maps, hyperspectral airborne and multispectral satellite images, and geophysical images to databases in many formats.
- Pipelines, electric lines, roads, ramps, and other mining facilities change frequently. Engineers and operations staff use GIS for facility planning applications.
- Keeping track of existing infrastructure and integrating it with the mine plan and block models can be achieved with GIS.
- GIS can also be used to integrate recent survey data with block models or mine design data from other mining software packages such as GeoSoft, Vulcan, MineSight, SURPAC Range, or Mining Visualization System (MVS).

British Columbia Administers Mining Titles Online

MTO is the first e-commerce, GIS Web-enabled system for mineral title acquisition in British Columbia.

With this system, members of the mineral exploration industry, authorized agents, and staff from mining companies can acquire mining rights by selecting a claim on an electronic map rather than staking a claim on the ground.

Establishing secure title on an accurate digital map base integrated with other online resources has streamlined the entire claim acquisition process.

[About](#)[Debug](#)[Layers](#)[Legend](#)[Find Location](#)[Select](#)[Refresh Map](#)[Disclaimer](#)**Map Layers**

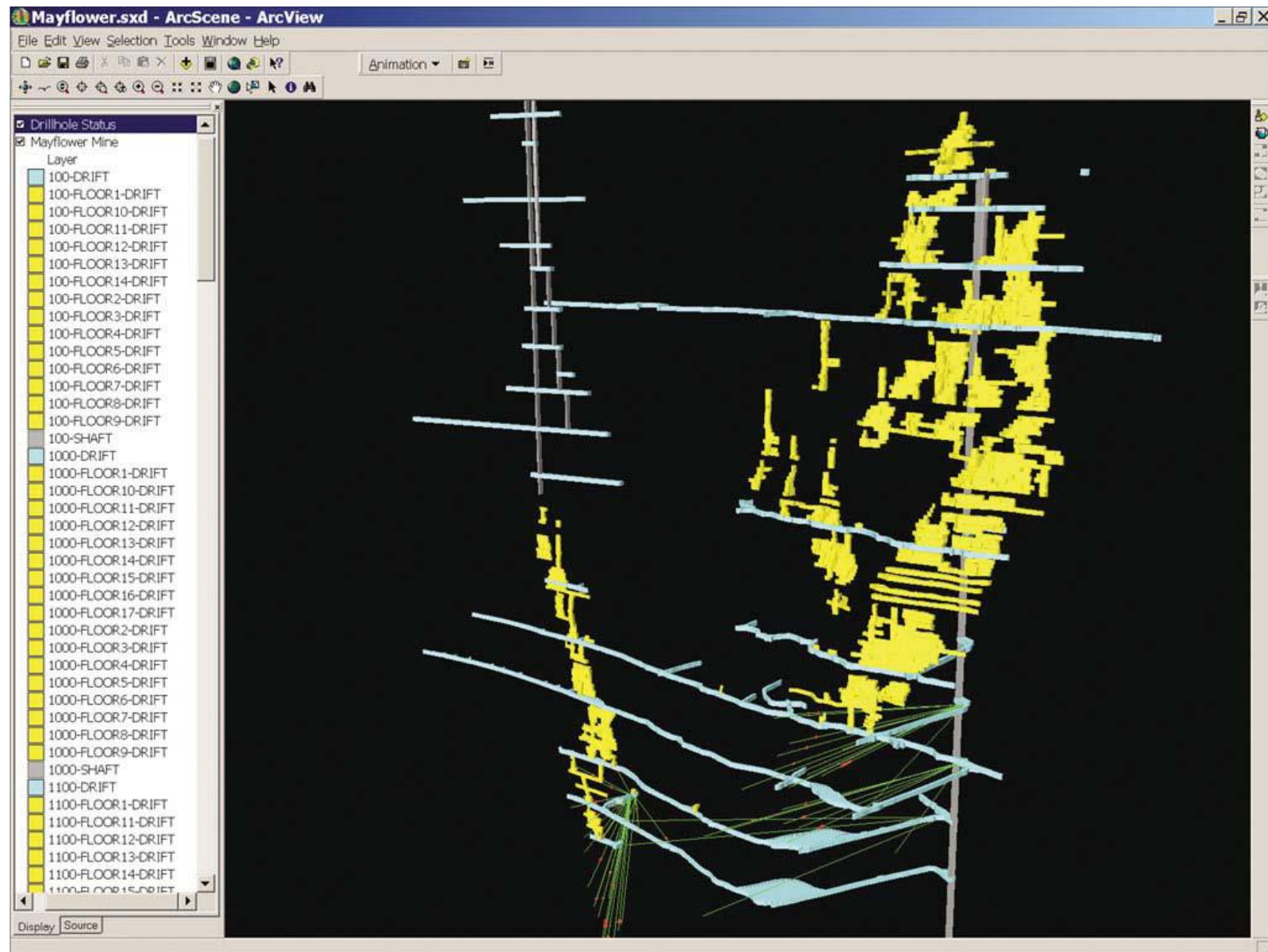
-  Selection Test
 - ☒  MTA Spatial Grid
-  other layers
 - ☒  Mining Division
 - ☒  Reserves (Sites)
 - ☒  Mineral Tenures
 - ☒  BC Boundary

[Refresh Map](#)**Notes:**

-  Click folder to show or hide contents.
- ☒ Click the check box to turn layer on and off.
-  This is the active identify layer.
-  Layer available for viewing. Click to make it the active identify layer.
-  Click to turn on all layers in the associated folder.
-  Click to turn off all layers in the associated folder.

Mining Gold in Montana With GIS and Underground Modeling

- In the mid-1990s, Brimstone Mining, Inc., purchased the historic Mayflower Mine in southwest Montana.
- The Mayflower Mine had been a top gold producer in the 1930s.
- The value of gold when Brimstone bought the mine, along with the prospect of applying more advanced mining techniques, provided the stimulus to reopen the mine.

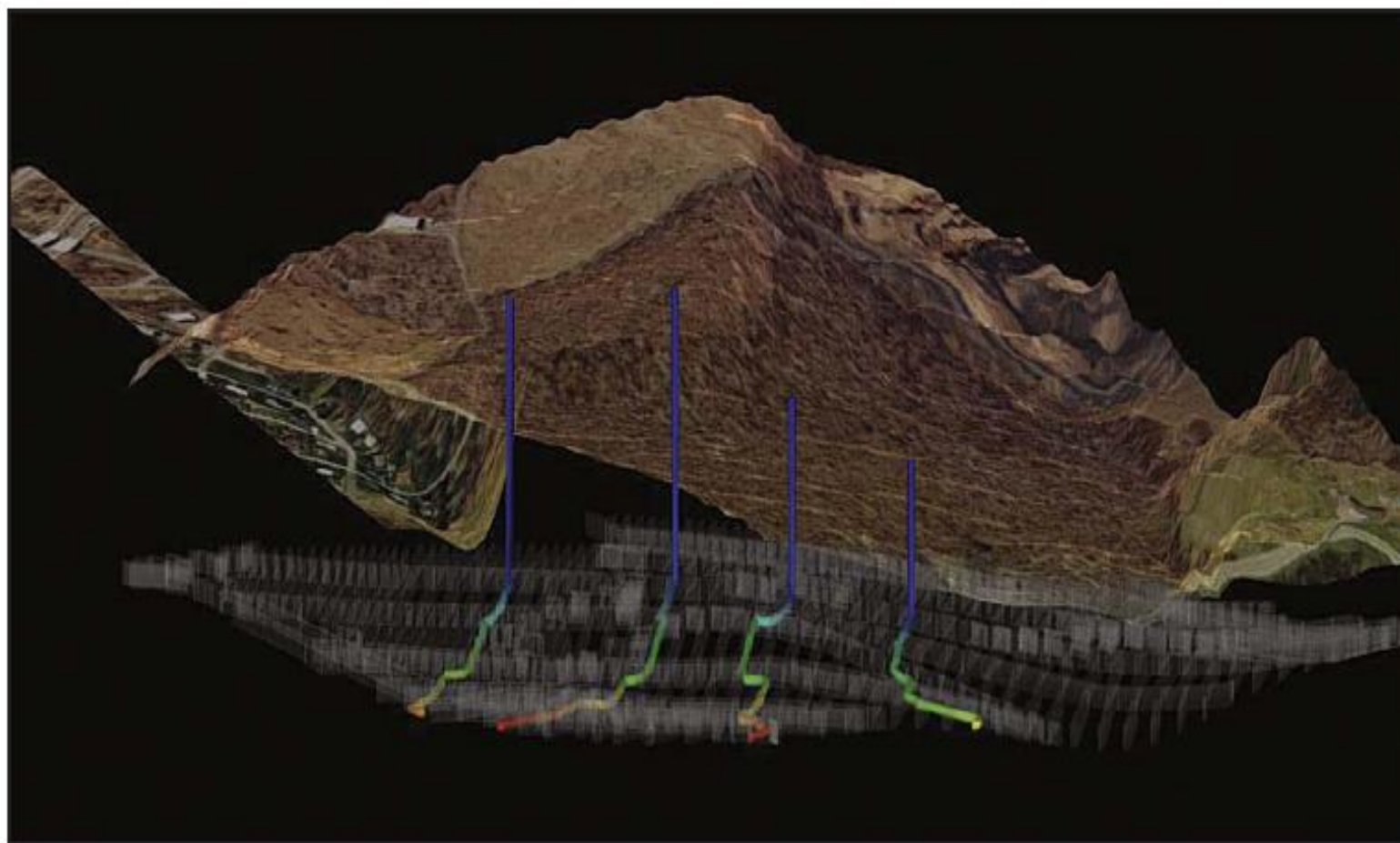


Stockton-on-Tees, United Kingdom, Identifies and Remediate Potentially Contaminated Land With GIS

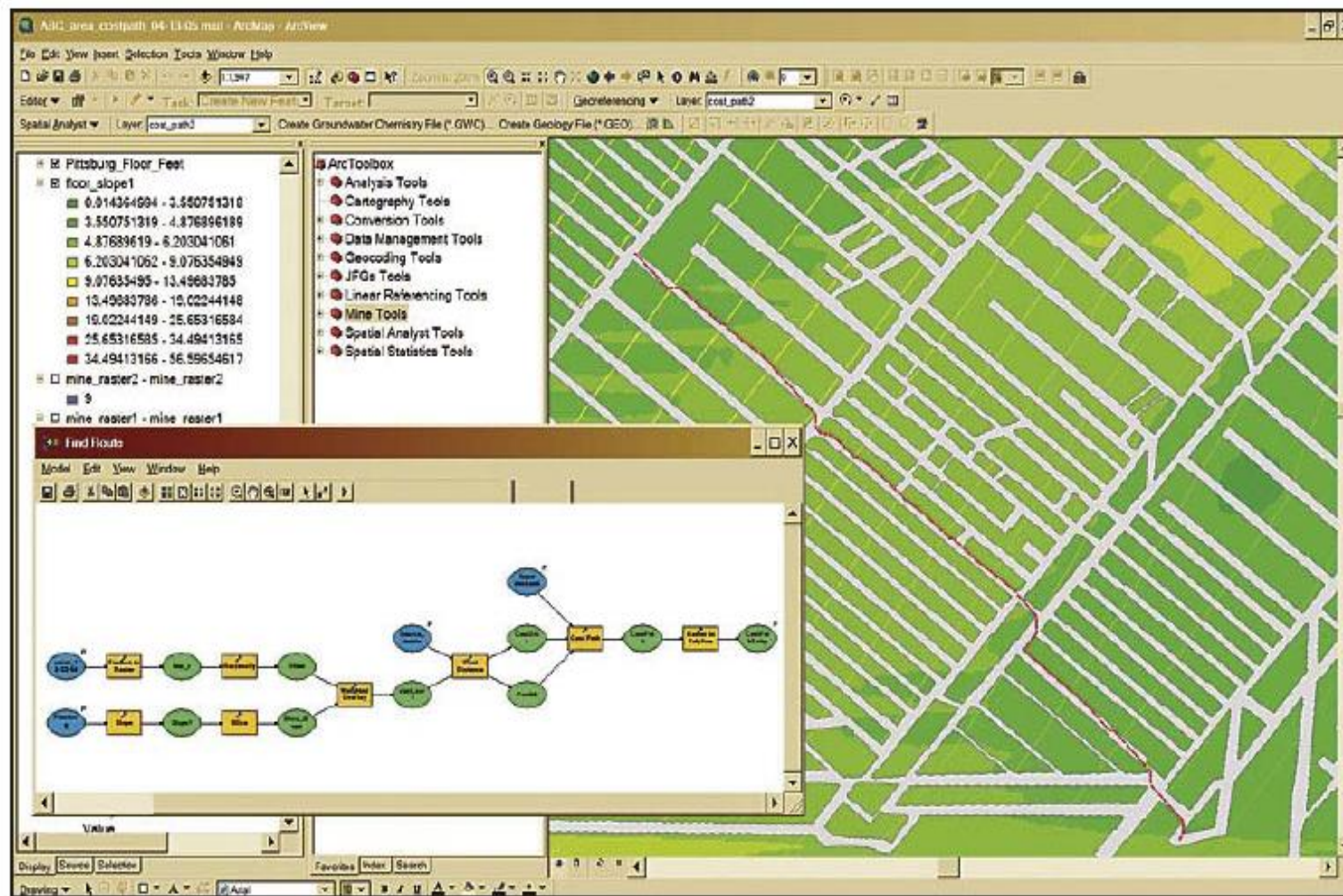
- A mixture of busy town centers, urban residential areas, and picturesque villages, Stockton-on-Tees Borough Council (SoTBC) lies in northeastern England.
- It began as an Anglo-Saxon settlement close to the River Tees.
- By the 17th century Stockton was established as the main port on the River Tees and had developed an important Baltic trade.
- It remained a largely agricultural district until the opening of the Stockton and Darlington Railway (the world's first public railway) in 1825, which brought about significant increases in the trade and population of Stockton.
- This began with the transport of lead from the dales and coal from local mines but later developed to the heavy industries that have left a considerable mark on the country today.
- These included mineral extraction, brick and tile manufacture, iron and steel manufacturing, shipbuilding, and engineering and chemical works.

In Maryland, Abandoned Coal Mine Stabilization Is Made Economical with GIS

- The Maryland Department of Natural Resources Power Plant Research Project (PPRP) was formed in 1971 to ensure that Maryland meets electricity demands at reasonable costs while protecting the state's valuable natural resources.
- A component of ensuring reasonable costs is the cost-efficient and environmentally friendly management of coal combustion by-products (CCPs) generated by coal-fired power plants in and around the state.
- As a legacy of coal mining, western Maryland is associated with detailed but often nongeoreferenced maps.
- The threat of land subsidence from mine tunnel (void) collapse and acid mine discharge (AMD) from these abandoned mines has adversely impacted development in and around increasingly sprawling communities.



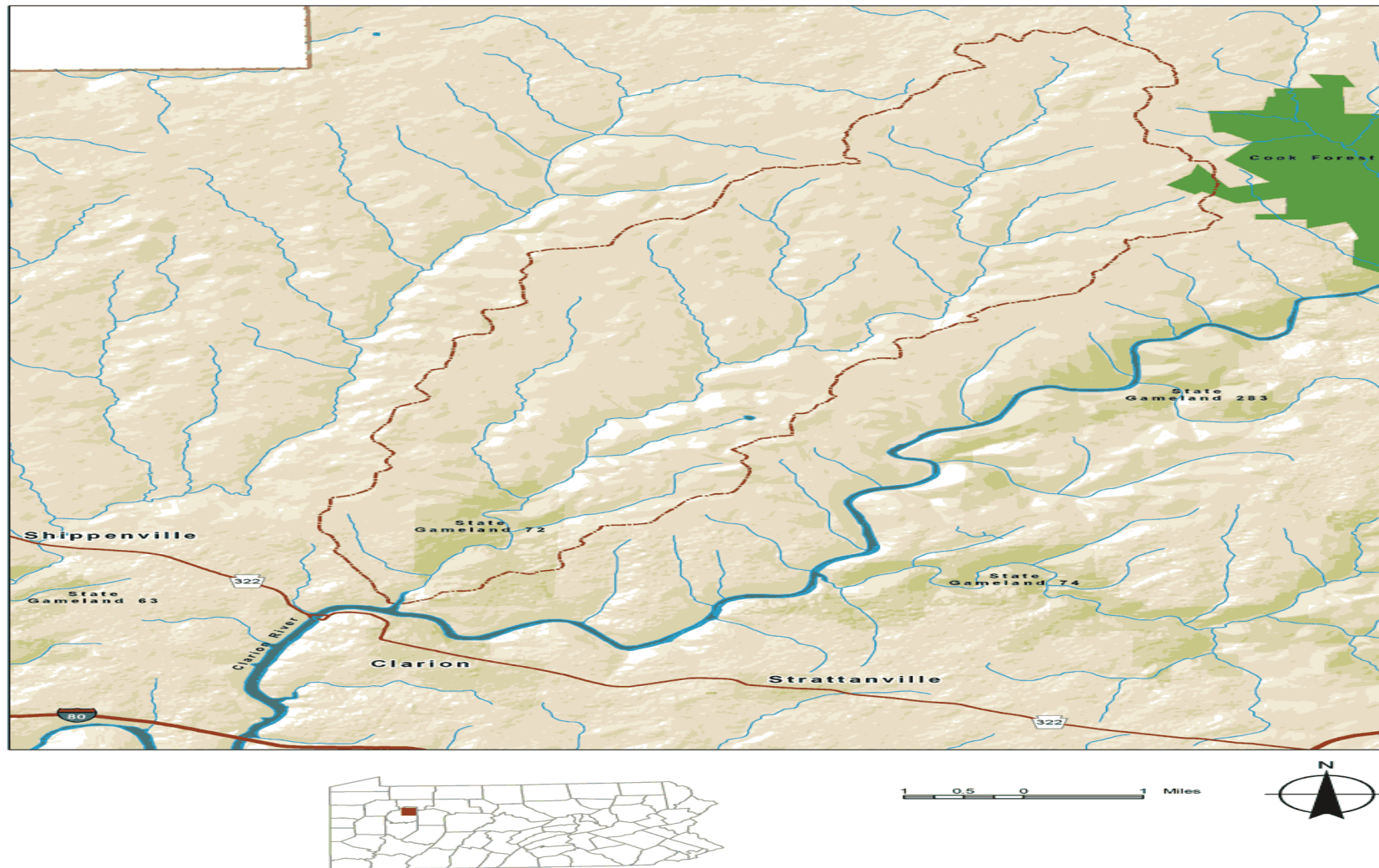
Simulated grout movement through the mine tunnel voids starting from land surface to the lowest point in mine tunnels.



GIS environment for estimating grout flow through mine tunnel voids.

Prioritizing Acid Mine Drainage Stream Remediation

- Coal mining activities in the Clarion River basin, located in northwestern Pennsylvania, have left scars on the landscape and severely polluted many streams in the basin.
- The area suffers from an environmental problem commonly associated with coal mining—acid mine drainage (AMD).
- The stream remediation process needs to be organized based on stream location, pollutant levels, and contaminant concentrations.
- GIS provided a means for bringing spatial and informational data together to simplify prioritization of the stream remediation process.
- The outcome was a series of maps summarizing different stream conditions that suggested an intervention plan based on the concentration levels of stream pollutant factors.



Streams are prioritized for remediation based on different combinations of acidity level and iron concentration